Module 3: Resampling and Simulation-Based Statistical Inference

MA 563 – Statistical Computation and Simulation

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$\Rightarrow$ $\mu, \sigma^2 \rightarrow F \rightarrow X_1 \cdots X_n \Rightarrow$ sample

$\mu = \rightarrow \hat{\theta}_1 = \bar{X} \rightarrow \leftrightarrow se(\bar{X}) \Rightarrow (\bar{X} \pm 2\, se(\bar{X}))$

$\hat{\theta}_2 = \text{med}(X_1 \cdots X_n)$

$\hat{\theta}_3 = \text{mod}(X_1 \cdots X_n)$

↳ First way → find analytical sampling dist^n of estimator $\hat{\theta}$

↳ → $X_1^1 \cdots X_n^1 \rightarrow \hat{\theta}_1^1 \rightarrow \bar{X}_1$ (e.g.)

we don't know this → $X_1^2 \cdots X_n^2 \rightarrow \hat{\theta}^2 \rightarrow \bar{X}_2$ → dist^n → Histogram

$X_1^N \cdots X_n^N \rightarrow \hat{\theta}^N \rightarrow \bar{X}_N$

sampling dist^n of $\bar{X}$, $\hat{\theta}$

→ $\hat{\theta}^{\bar{X}}$

Assumption $X_1 \cdots X_n \sim N(\mu, \sigma^2) \Rightarrow \mu, \sigma^2$

$\hat{\theta}_1 = \bar{X}_n \sim N(\mu, \frac{\sigma^2}{n})$

Sampling dist^n of $\bar{X}_n$

↳ $\bar{X}, \frac{S^2}{n}$, $S^2 = \frac{1}{n}\sum_{i=1}^{n}(X_i - \bar{X})^2$

↳ estimate $\bar{X}$

95% CI: $\bar{X} \pm 2S/\sqrt{n} \Rightarrow \frac{1}{n}\sum_{i=1}^{n} X_i + 2\frac{\sqrt{\frac{1}{n}\sum_{i=1}^{n}(X_i - \bar{X})^2}}{n}$

$\sigma^2 < \infty$

$X_1 \cdots X_n \sim F(\mu, \sigma^2)$.

$\Rightarrow \bar{X}_n \sim N(\mu, \sigma^2/n) \Rightarrow n \rightarrow \infty$

→ Sample median → , $X_1 - - X_n \sim F(\mu, \sigma^2)$

$$\hat{\theta}_2 = \text{med}(X_1 - - X_n)$$

$$\hat{\theta}_2 \sim N\left[\mu, \frac{1}{4[f'(\mu)]^2 n}\right]$$

f : pdf

→ sample mode → $\hat{\theta}_3 = \text{mod}(X_1 - - X_n)$

$\hat{\theta}_3 \sim$?

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Module 3 at a Glance

Module 3: Resampling and Simulation-Based Statistical Inference (10 Lectures)

Bootstrap principles, parametric vs nonparametric bootstrap, cross-validation, jackknife (brief), simulation-based uncertainty quantification, computational trade-offs, likelihood-based inference via simulation, parametric simulation for inference, confidence intervals via simulation, hypothesis testing using Monte Carlo, permutation tests, bootstrap-based inference (deep dive), simulation vs asymptotic inference.

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How Module 3 builds on Modules 1 and 2

- **Module 1** gave us generators: we can draw from any F we can write down.
- **Module 2** gave us Monte Carlo: any expectation can be approximated by an average.
- **Module 3** applies both to *statistical inference itself*: standard errors, confidence intervals, p-values and prediction error – computed by simulation instead of by formula.

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Every Estimate Needs an Error Bar

- We observe data $x_1, \ldots, x_n \overset{\text{iid}}{\sim} F$ and compute a statistic $\hat{\theta} = s(x_1, \ldots, x_n)$ to estimate a parameter $\theta = T(F)$.

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The central question of Module 3

The sampling distribution depends on the *unknown* F, and we see only *one* sample. Can we approximate it anyway – for *any* statistic, without deriving a formula?

The Textbook Case – and Why It Is Special

- For the sample mean $\bar{X}$, classical theory delivers everything:

$$\mathrm{se}(\bar{X}) = \frac{\sigma}{\sqrt{n}} \approx \frac{s}{\sqrt{n}}, \qquad \bar{X} \pm t_{n-1,\,\alpha/2}\,\frac{s}{\sqrt{n}}.$$

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The honest picture

Closed-form standard errors exist for a small, special family of statistics. Real analyses use statistics that were chosen for their *scientific* meaning, not for having a tidy variance formula.

Real Problems with No Standard-Error Formula

Medicine: median survival

A small trial ($n = 25$) reports *median* time to recovery. Patients are expensive; the data are skewed. How uncertain is the median?

$\text{med}(X_1, \ldots, X_n)$ = median time of recovery

= 5 days

$\Rightarrow$ (4, 6) $\rightarrow$ 95% CI

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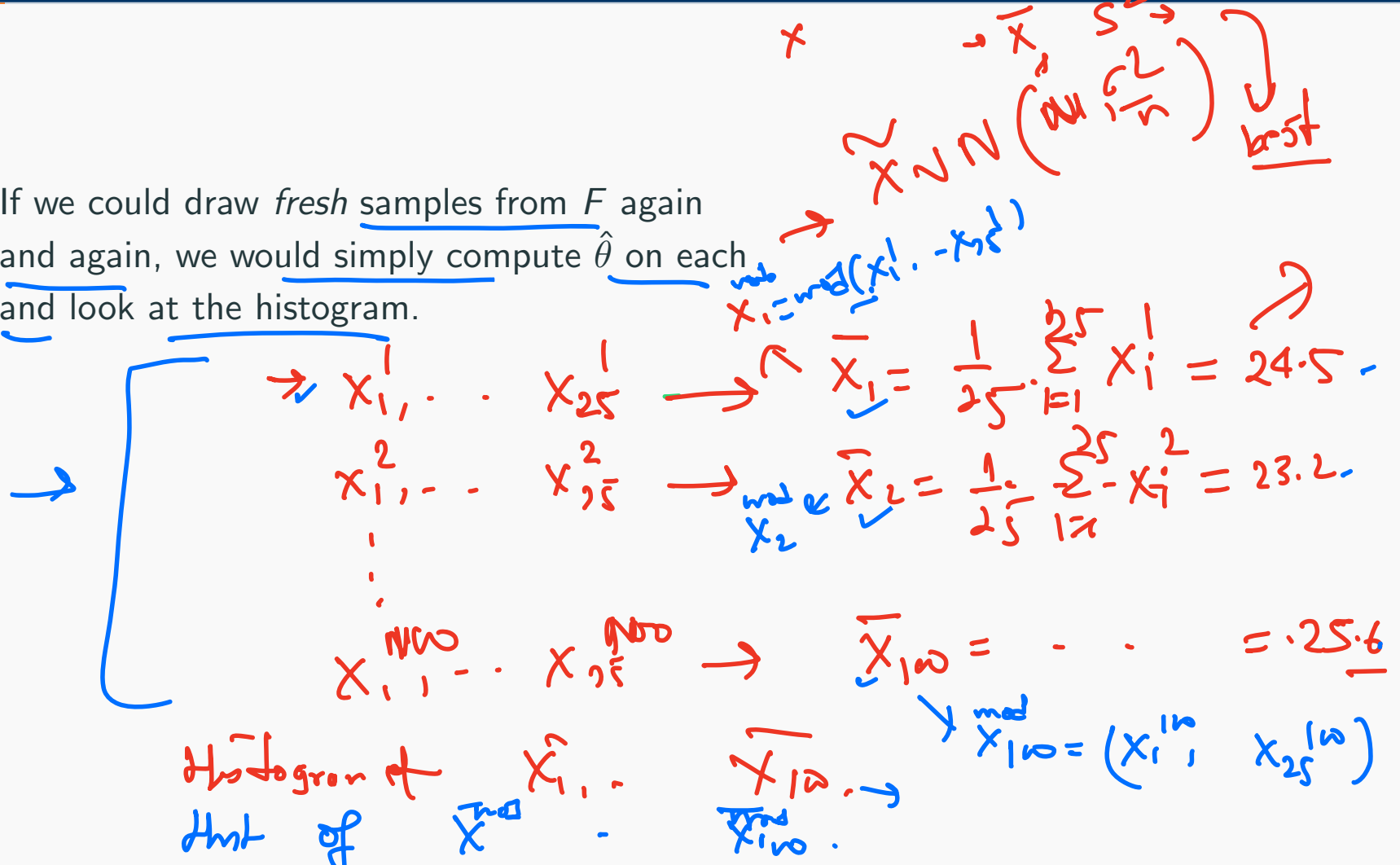
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The common thread

Each asks for a sampling distribution that classical theory cannot supply cleanly. Each is answered in practice by **resampling** or **simulation**.

What We Would Do If We Could Repeat the Experiment

- If we could draw *fresh* samples from F again and again, we would simply compute $\hat{\theta}$ on each and look at the histogram.



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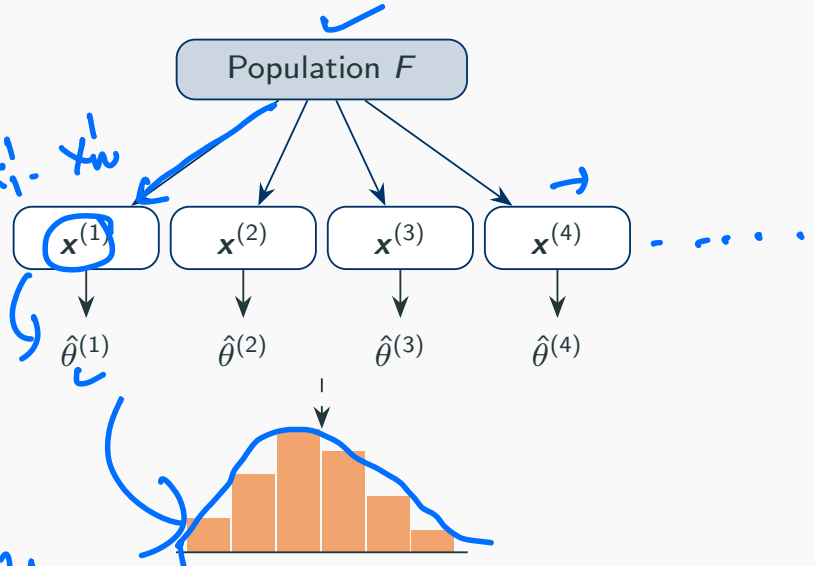
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Parametric Bootstrap

Assumption on F → parametric case → F: $N(\mu, \sigma^2)$, Gamma(α, β) e.g.

estimating F $\Leftrightarrow$ estimating unknown parameter

$\Leftrightarrow$ μ, σ^2 estimating for $F = N(\mu, \sigma^2)$

↓

$\hat{\mu}_{MLE}, \hat{\sigma}^2_{MLE}$

Sample $(x_1 \dots x_n)$

estimate F

estimate $F \rightarrow \hat{F}(\hat{\mu}_{MLE}, \hat{\sigma}^2_{MLE})$

↓ Exactly know what is F

use rnorm 1 → to simulate for Normal distribution with mean $\hat{\mu}_{MLE}$, $\hat{\sigma}^2_{MLE}$

→ Sample from $\hat{F}(\hat{\mu}, \hat{\sigma}^2)$

→ $x_1^1 \dots x_n^1 \rightarrow \hat{\theta}_1$

$x_1^2 \dots x_n^2 \rightarrow \hat{\theta}_2$ ⟷ Histogram of sampling distr of $\hat{\theta}$

$\vdots$

$x_B^1 \dots x_B^2 \rightarrow \hat{\theta}_B$

F: Non parametric

→ can't be written in term of finite no of parameter

estimate F. by empirical cdf →

$$P(X \le t) = \hat{F}_n(t) = \frac{1}{n} \cdot \sum_{i=1}^{n} I(X_i \le t), \quad X_1 \dots X_n \text{ sample}$$

↳ estimators

$$\Rightarrow \hat{F}_n(-\infty) = \frac{1}{n} \cdot \sum_{i=1}^{n} I(X_i \le -\infty) = 0$$

$$\hat{F}_n(\infty) = \frac{1}{n} \sum_{i=1}^{n} I(X_i \le \infty) = \frac{n}{n} = 1$$

$$\text{if } t_1 < t_2 \Rightarrow \hat{F}_n(t_1) \le \hat{F}_n(t_2)$$

$$\frac{1}{n} \sum_{i=1}^{n} I(X_i \le t_1) \le \frac{1}{n} \cdot \sum_{i=1}^{n} I(X_i \le t_2)$$

$$X_i \le t_1 \Rightarrow X_i \le t_2$$

$$\frac{\sum_{i=1}^{n} I(X_i \le t_1)}{n} \le \frac{\sum_{i=1}^{n} I(X_i \le t_2)}{n}$$

↳ $\hat{F}_n(t_1) \le \hat{F}_n(t_2)$ → non-parametric

Empirical CDF ⇒

$$\boxed{\hat{F}_n(t) = \frac{1}{n} \cdot \sum_{i=1}^{n} I(X_i \le t)}$$

↳ an estimate of F in case of non-parametric

Now we can simulate sample from $\hat{F}_n(t)$

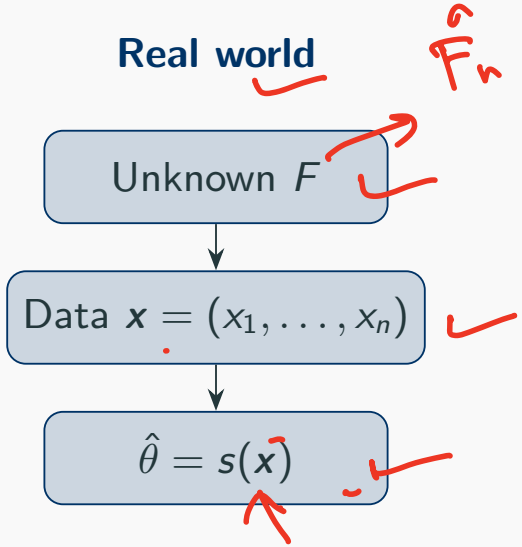
$\Rightarrow$ $X_1^1 \; \cdots \; X_n^1 \sim \hat{F}_n(t) \longrightarrow \hat{\theta}_n^1$

$\vdots$

$X_1^B \; \cdots \; X_n^B \sim \hat{F}_n(t) \longrightarrow \hat{\theta}_n^B$

Histogram of it $\rightarrow$ sampling dist^n of $\hat{\theta}$.

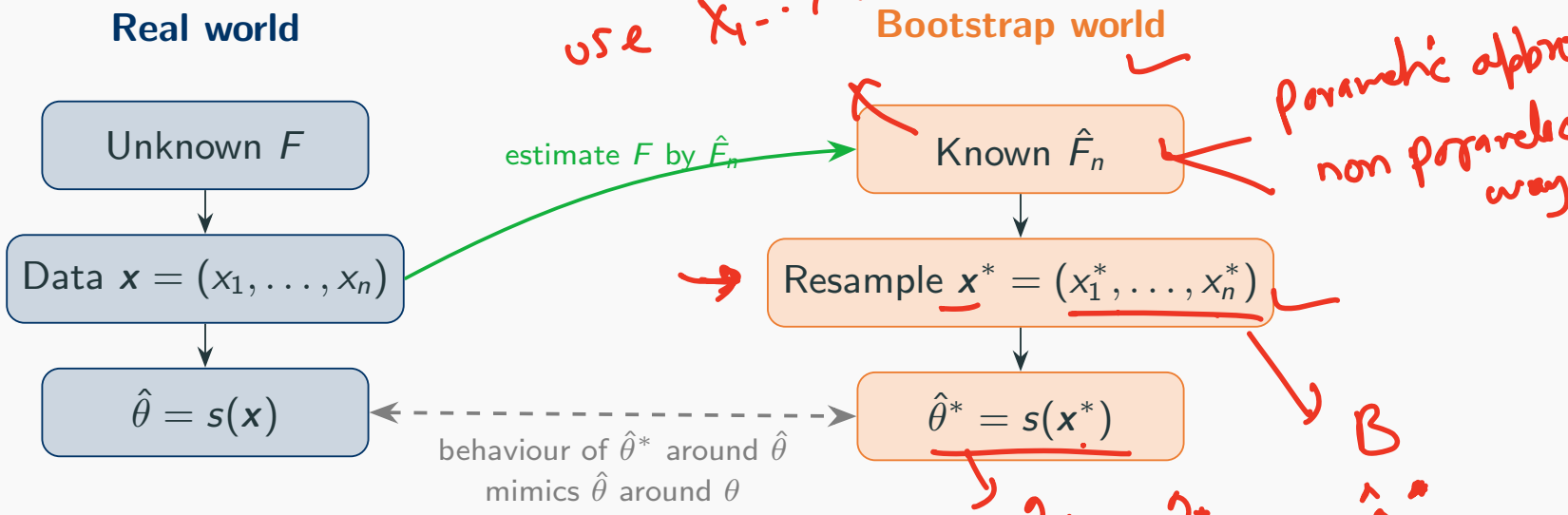
The Resampling Idea: Let the Sample Stand In for the Population



Efron (1979): the bootstrap

Everything we cannot compute in the real world (because F is unknown) we **can** compute in the bootstrap world (because $\hat{F}_n$ is known) – by simulation.

The Resampling Idea: Let the Sample Stand In for the Population



use $x_1 \ldots x_n$ estimate F

Parametric approach / non parametric way

B

$\hat{\theta}_1^*, \hat{\theta}_2^* \ldots \hat{\theta}_B^*$

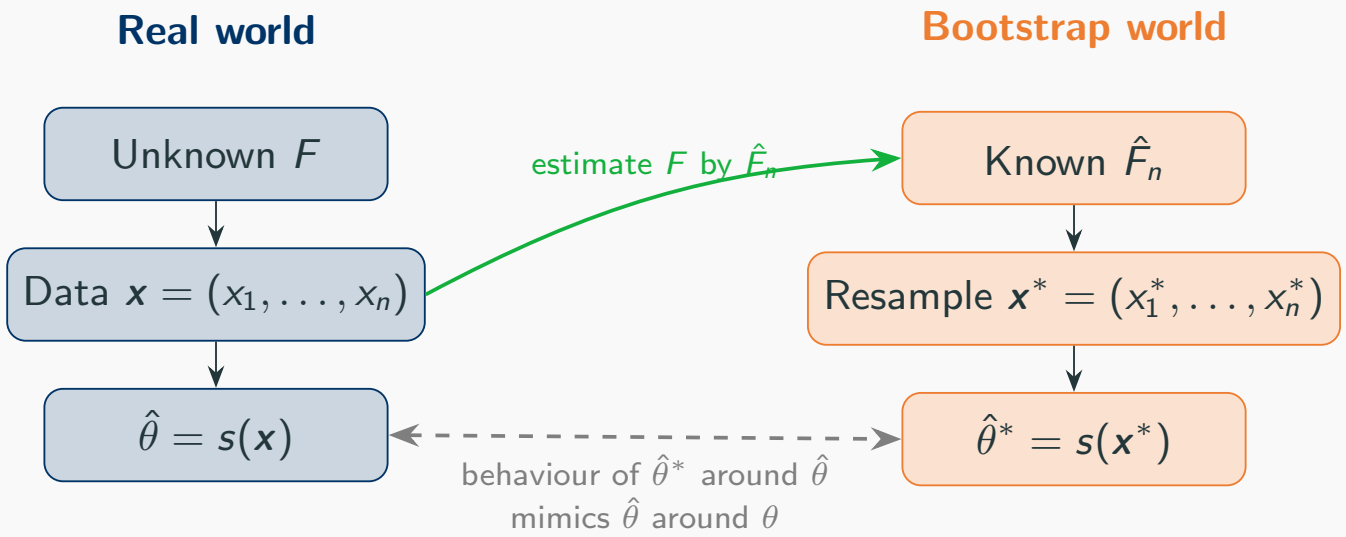
CI → $\bar{\theta}^* \pm 2\,se$

$\bar{\theta}^* = \frac{1}{B}\sum_{i=1}^{B} \hat{\theta}_i^*$

$se = \sqrt{\frac{1}{B}\sum_{i=1}^{B}(\hat{\theta}_i^* - \bar{\theta}^*)^2}$

Histogram of it will give you sampling distn

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The Empirical Distribution Function

Non parametric approach.

Definition

The empirical distribution $\hat{F}_n$ puts probability mass $1/n$ on each observation:

$$\hat{F}_n(t) = \frac{1}{n} \sum_{i=1}^{n} \mathbf{1}\{x_i \leq t\}.$$

indicator fn

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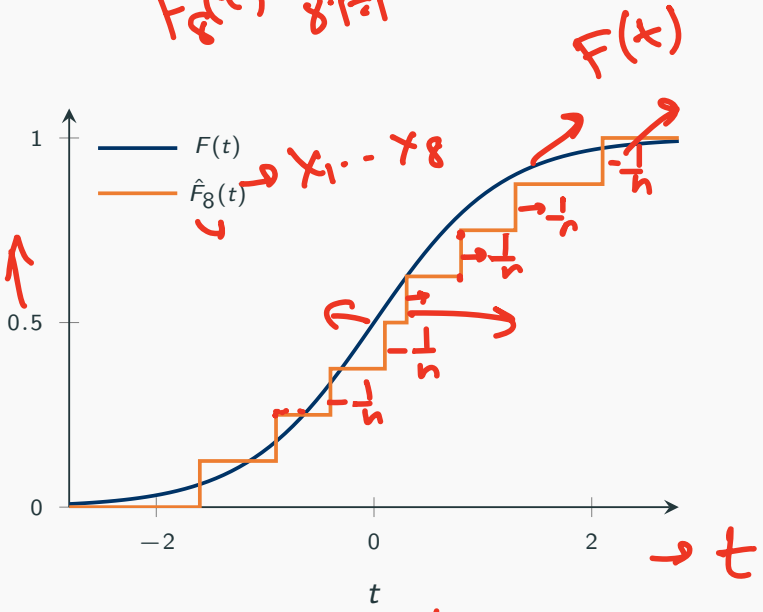
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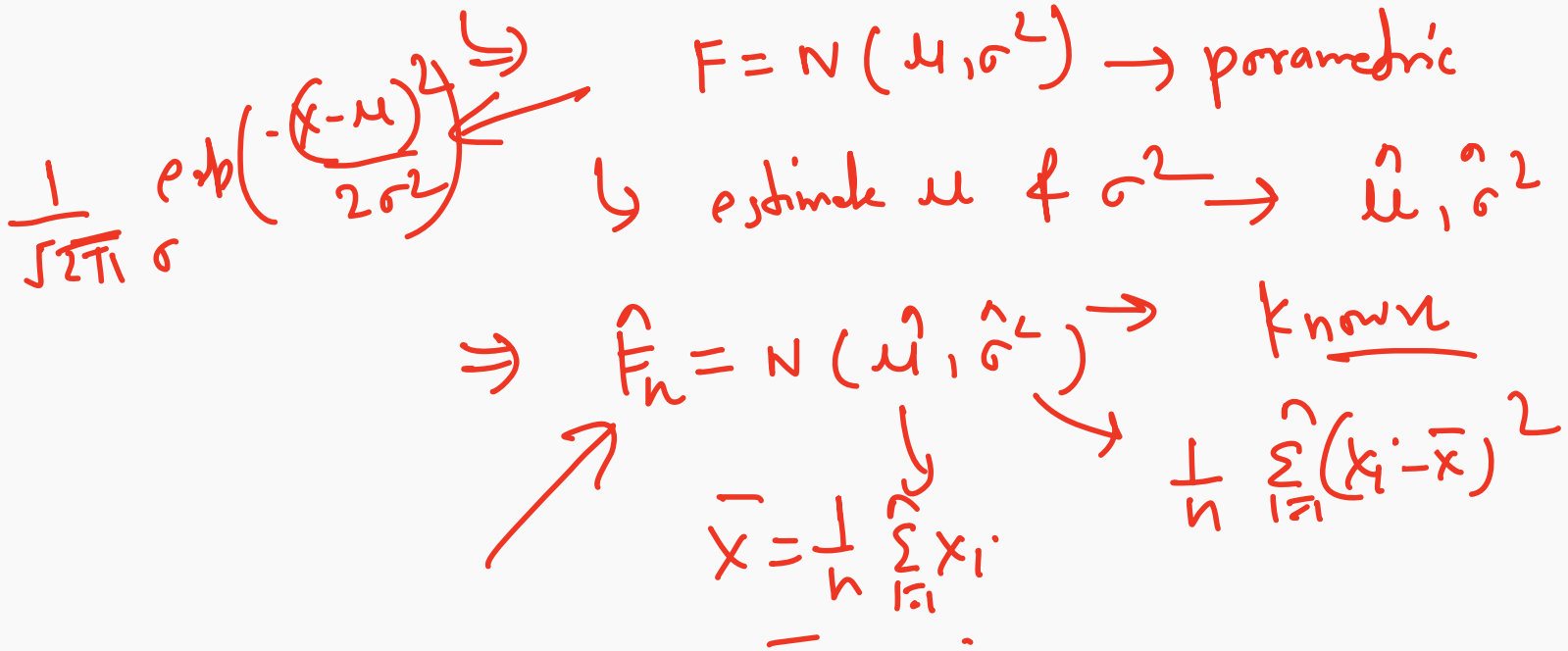
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The Plug-in Principle

- Many parameters are *functionals* of F: $\theta = T(F)$. The **plug-in estimator** is $\hat{\theta} = T(\hat{F}_n)$.



The Plug-in Principle

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Parameter	$T(F)$	Plug-in $T(\hat{F}_n)$
Mean	$\int x\, dF(x)$	$\bar{x}$
Variance	$\int (x-\mu)^2 dF(x)$	$\hat{\sigma}^2 = \frac{1}{n}\sum (x_i - \bar{x})^2$ (divisor n!)
Median	$F^{-1}(1/2)$	sample median
Correlation	$\mathrm{Cov}(X,Y)/(\sigma_X \sigma_Y)$	sample correlation r

CDF

$\frac{dF(x)}{dx} = f(x)$ → PDF → $E(X) = \int x f(x)\, dx$

$dF(x) = f(x)\, dx$

$f(x)dx = dF(x)$ CDF

$V(X) = \int (x-\mu)^2 f(x) dx$ $dF(x)$

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The bootstrap is the plug-in principle applied one level up

The standard error is *also* a function of F: $\mathrm{se}_F(\hat{\theta})$. Its plug-in estimate is

$$\mathrm{se}_{\hat{F}_n}(\hat{\theta}^*) = \text{standard deviation of } s(\boldsymbol{x}^*) \text{ when } \boldsymbol{x}^* \overset{\text{iid}}{\sim} \hat{F}_n.$$

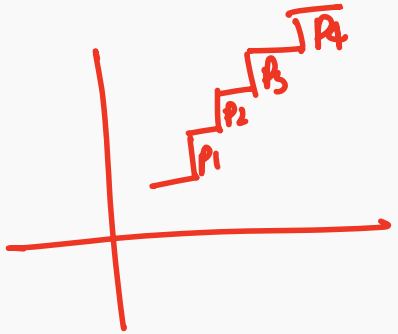
This is the **ideal bootstrap estimate** of the standard error.

$$\mathrm{se}_{\hat{F}_n}(\hat{\theta}^*) = \sqrt{\frac{1}{B}\sum_{i=1}^{B} (\hat{\theta}_i^* - \bar{\theta}^*)^2}$$

Sampling from $\hat{F}_n$ = Resampling With Replacement

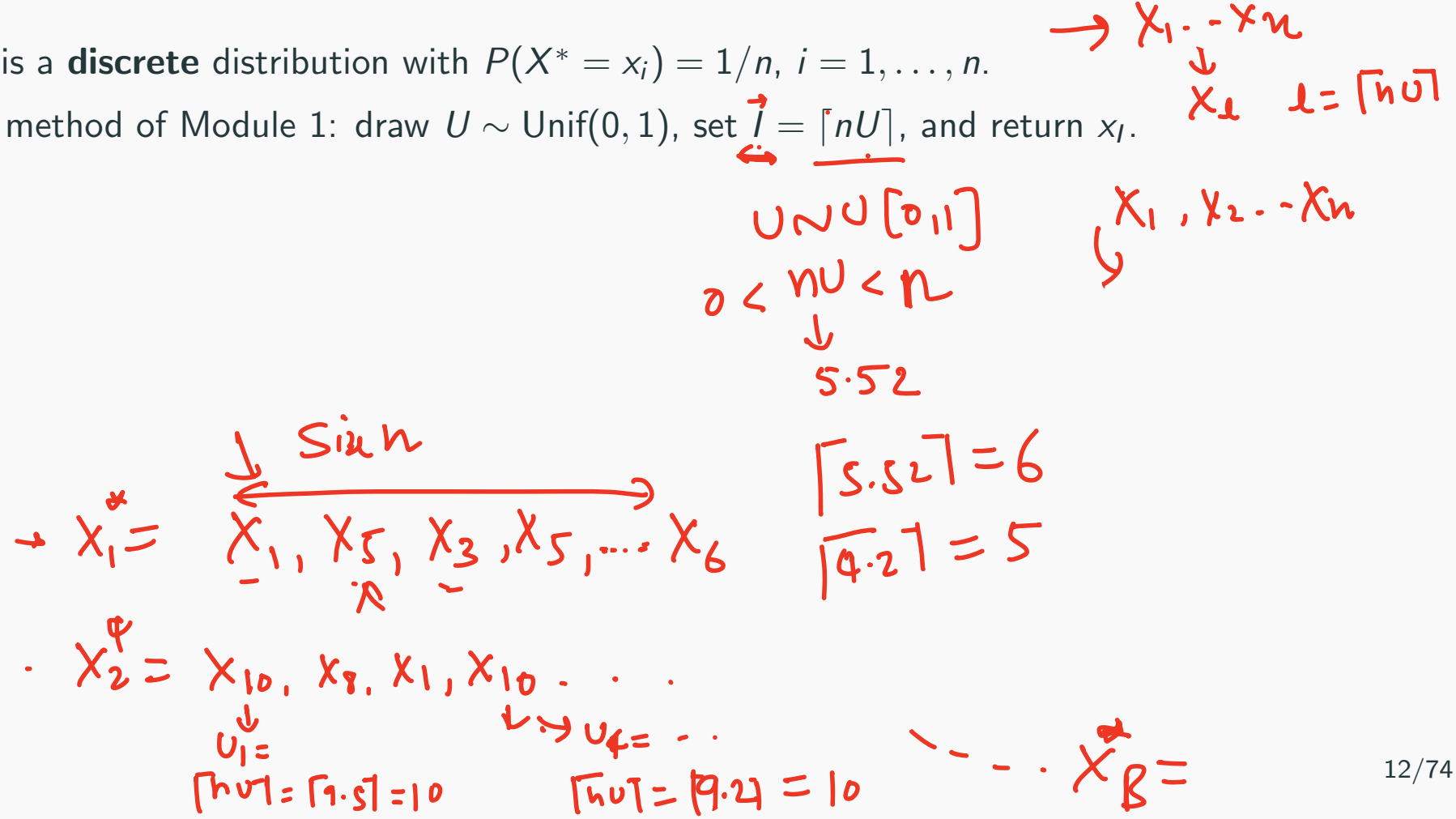
- $\hat{F}_n$ is a **discrete** distribution with $P(X^* = x_i) = 1/n$, $i = 1, \ldots, n$.

$p(x = x_i) = p_i$
$= \frac{1}{n}$



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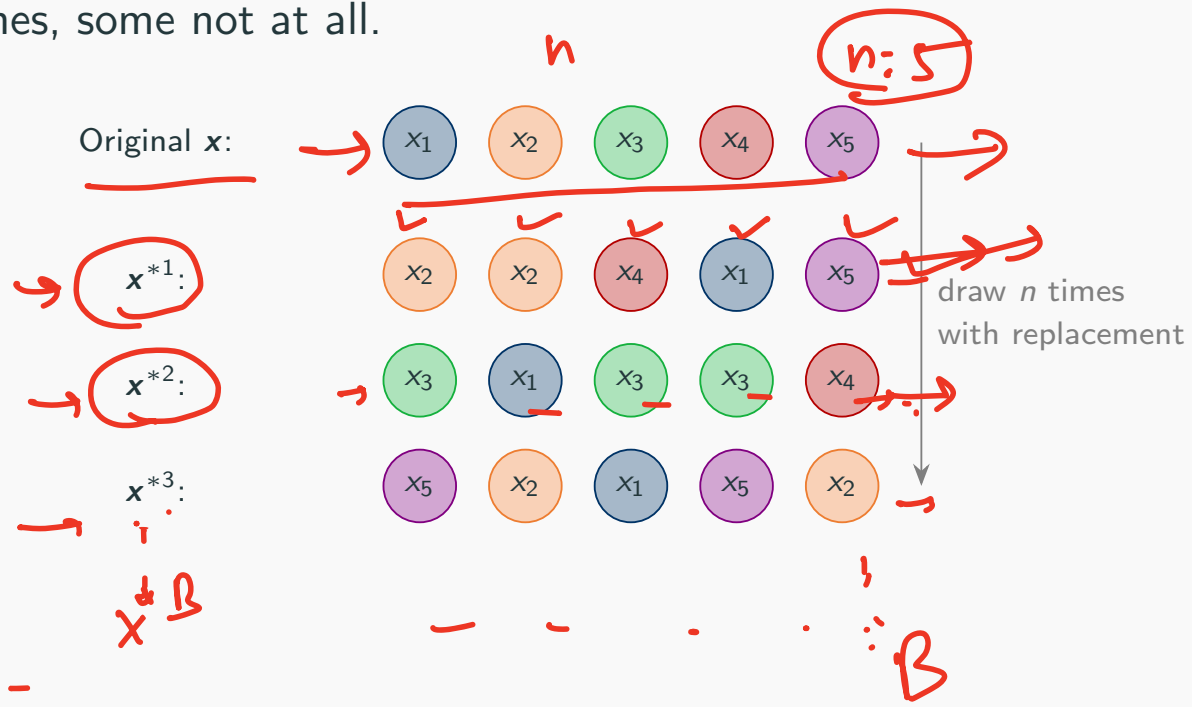
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- Repeating n times gives a bootstrap sample $\boldsymbol{x}^* = (x_{I_1}, \ldots, x_{I_n})$ – some observations appear several times, some not at all.

$x_1 \cdots x_n \rightarrow$ distinct

$\boldsymbol{x}_1^* = (x_{11} \cdots x_{1n})$ —

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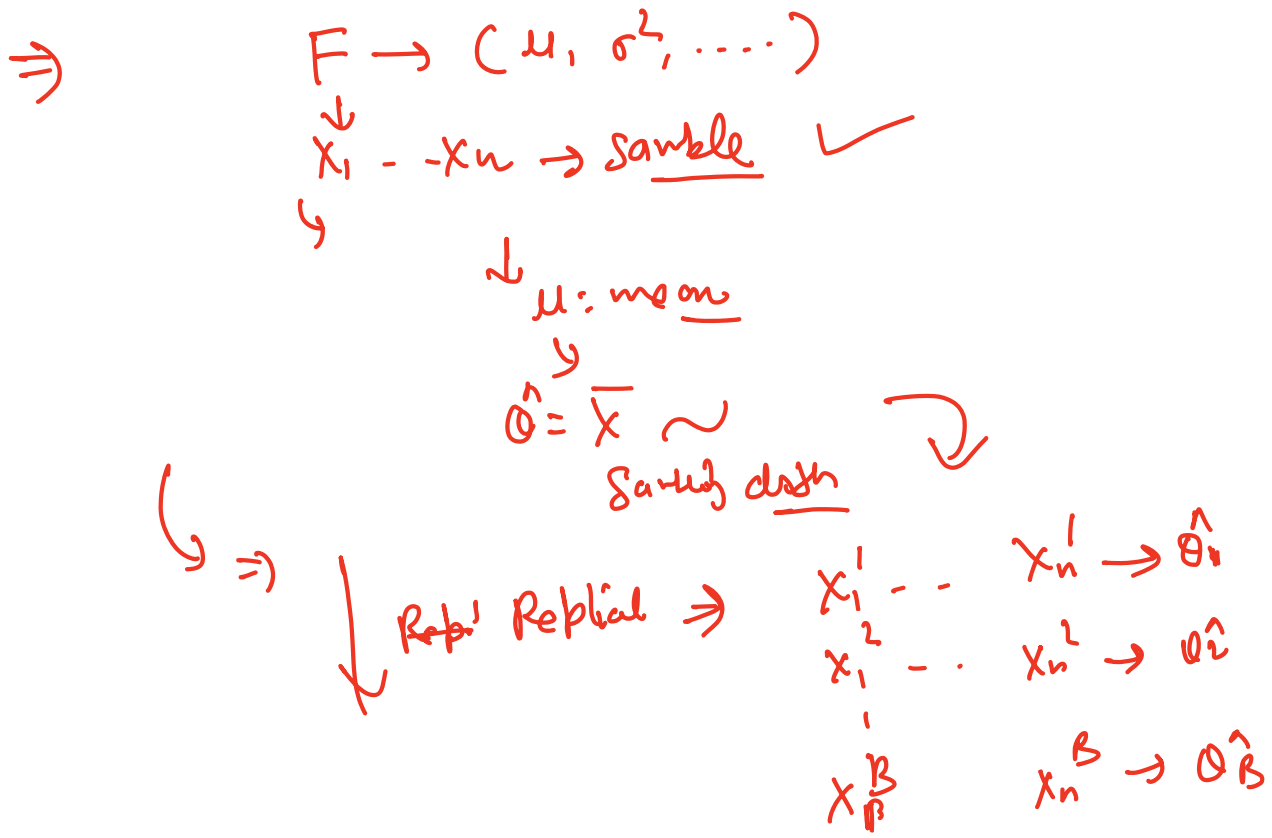


The Nonparametric Bootstrap: Algorithm

Algorithm: Nonparametric Bootstrap

Given data $\boldsymbol{x} = (x_1, \ldots, x_n)$ and a statistic $s(\cdot)$, with $\hat{\theta} = s(\boldsymbol{x})$:

1. For $b = 1, \ldots, B$:
   (a) Draw $\boldsymbol{x}^{*b} = (x_1^{*b}, \ldots, x_n^{*b})$ by sampling n values **with replacement** from $\boldsymbol{x}$.
   (b) Compute the bootstrap replication $\hat{\theta}_b^* = s(\boldsymbol{x}^{*b})$.
2. Use the empirical distribution of $\hat{\theta}_1^*, \ldots, \hat{\theta}_B^*$ as an approximation to the sampling distribution of $\hat{\theta}$.


F → (μ, σ², ...)
X1 - - Xn → sample
μ : mean
θ̂ = X̄ ~
Sampling distn
Repl' Replicat ⇒
X1¹ - - Xn¹ → θ̂1
X1² - - Xn² → θ̂2
X1^B Xn^B → θ̂B

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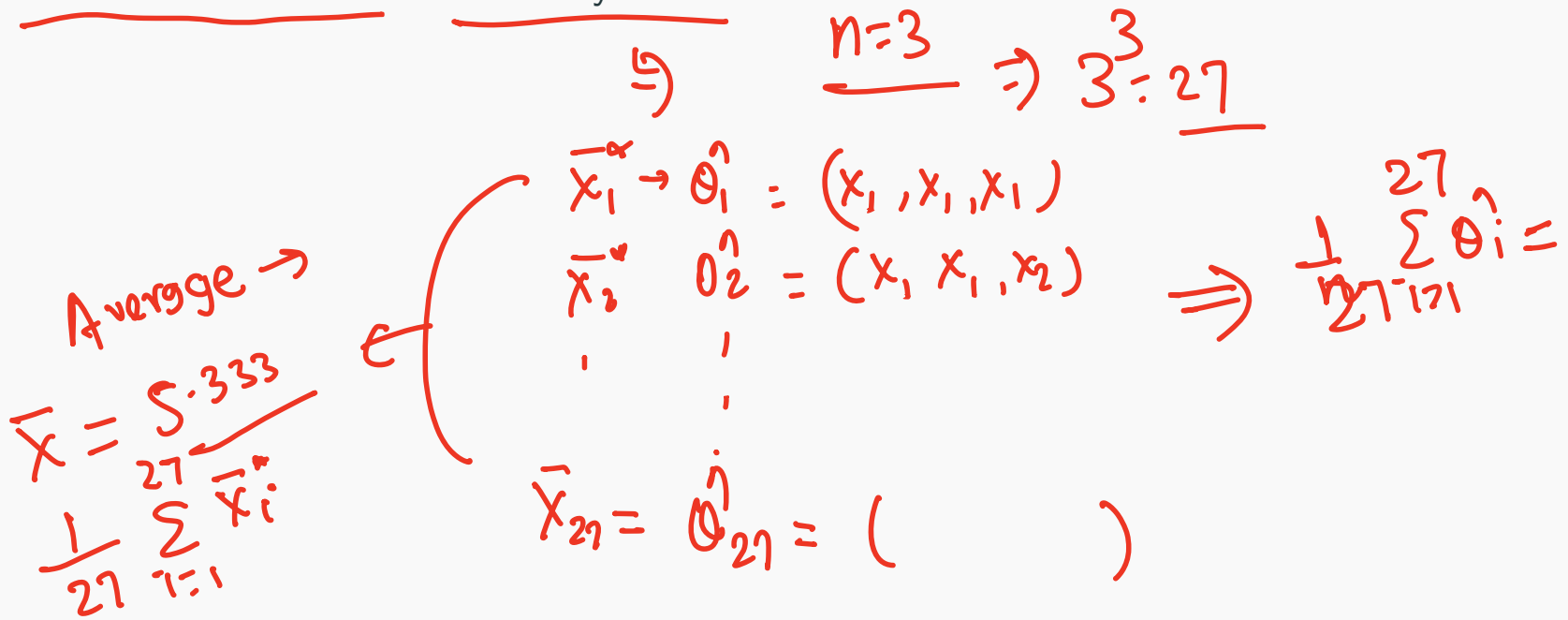
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- Step 2 is precisely **crude Monte Carlo** (Module 2) applied to the bootstrap world.

A Toy Example by Hand: $x = (2, 5, 9)$

- $\bar{x} = 16/3 \approx 5.333$. There are $3^3 = 27$ equally likely ordered resamples, but only $\binom{5}{3} = 10$ distinct ones (as multisets), e.g. $(2, 2, 2)$, $(2, 5, 9)$, $(9, 9, 5)$, ...

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$$\mathrm{se}_{\hat{F}_n}(\bar{x}^*) = \frac{\hat{\sigma}}{\sqrt{n}} = \sqrt{\frac{8.222}{3}} \approx 1.656,$$

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Compare with the textbook

$s/\sqrt{n} = \sqrt{12.333/3} \approx 2.028.$

$$\frac{(x_1-\bar{x})^2 + (x_2-\bar{x})^2 + (x_3-\bar{x})^2}{2}$$

$n-1$

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Lesson: the bootstrap uses plug-in (divisor-n) variability, so it slightly *underestimates* spread for very small n – the factor $\sqrt{(n-1)/n} \to 1$ quickly.

$n = \text{one}$
$\Rightarrow$ cloud $\Leftrightarrow$ bootstrap world $\rightarrow$

How Big Is the Bootstrap World?

- The number of distinct bootstrap samples of size n is $\binom{2n-1}{n}$:

$n = 3$	$n = 10$	$n = 20$
10	92,378	$\approx 6.9 \times 10^{10}$

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Two separate layers of approximation

$$\underbrace{\mathrm{se}_F(\hat{\theta})}_{\text{truth}} \overset{\text{statistical error}}{\approx} \underbrace{\mathrm{se}_{\hat{F}_n}(\hat{\theta}^*)}_{\text{ideal bootstrap}} \overset{\text{Monte Carlo error}}{\approx} \underbrace{\widehat{\mathrm{se}}_B}_{\text{what we compute}}$$

Increasing B removes only the second error. The first is fixed by the data (it shrinks only with n).